

GIT FUNDAMENTALS

Git Fundamentals

Version Control System

Track changes in code with history, enabling collaboration and reverting to previous states

Distributed Architecture

Works offline with full local repository; no constant network connection needed

Powerful Branching

Create isolated environments for feature development, fixes, and experiments

Lightweight

Small footprint with high performance even for large projects

Open Source

Free and widely supported across platforms

Multiple Remote Support

Can connect to multiple remote repositories simultaneously

Basic Operations

```
git status # Check status of working directory
git add <file> # Add file to staging area
git add . # Add all changes to staging area
git commit -m "Commit message" # Commit staged changes
git log # View commit history
git diff # Show changes between working directory and staging
git diff --staged # Show changes between staging and last commit
```

Core Git Concepts

Repository States

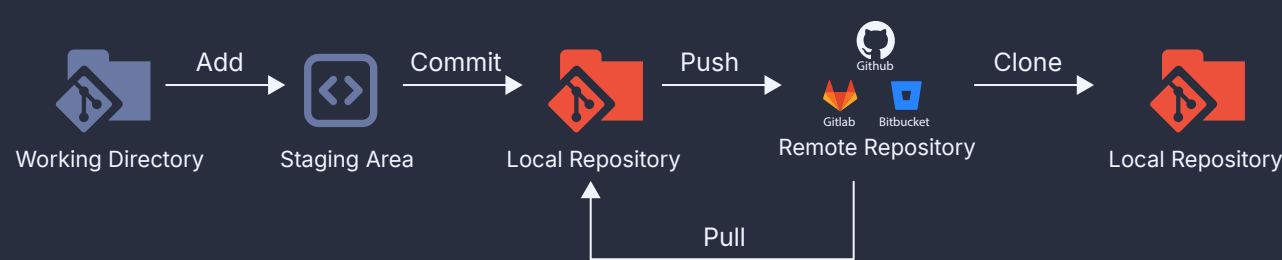
Working Directory:	Where you edit files
Staging Area:	Prepared changes for commit
Local Repository:	Committed history on your machine
Remote Repository:	Shared server repository (GitHub, GitLab, BitBucket, etc.)



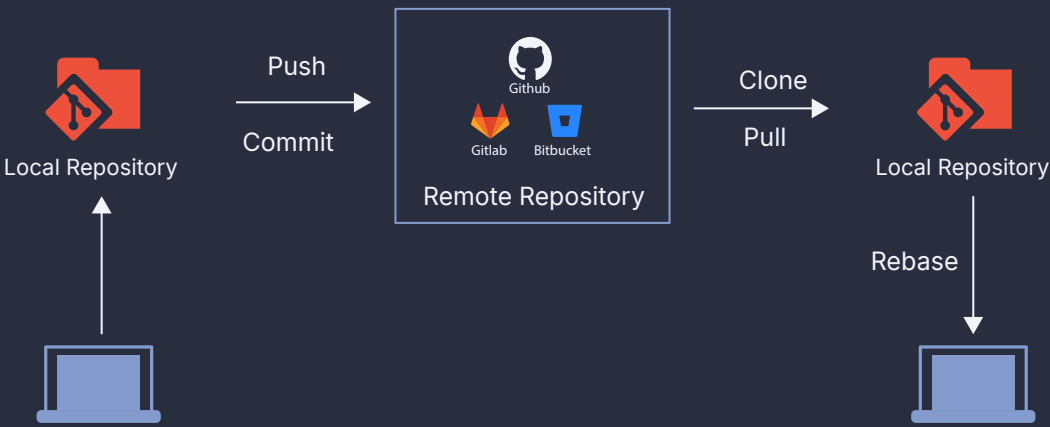
Branching & Merging

```
git branch # List branches
git branch <branch-name> # Create new branch
git checkout <branch-name> # Switch to branch
git checkout -b <branch-name> # Create and switch to new branch
git merge <branch-name> # Merge branch into current branch
git branch -d <branch-name> # Delete branch locally
```

Basic Workflow



Git Remote Repositories



Essential Git Commands

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
git init #Initialize new repository
git clone <repository-url> #Clone existing repository
```

Remote Operations

```
git remote -v # List remote repositories
git remote add <name> <url> # Add remote repository
git fetch <remote> # Download objects from remote
git pull <remote> <branch> # Fetch and merge changes
git push <remote> <branch> # Push commits to remote
git push -u <remote> <branch> # Push and set upstream
```